Eric Yang

eric.yang2@ucsf.edu | (310) 461-5746 | San Francisco, CA | Portfolio | Github | LinkedIn

Machine learning researcher with an M.S. in Health Data Science from UCSF focused on understanding what drives model behavior and whether empirical evaluations support the conclusions drawn from them. First-author experience designing controlled experiments, anatomical ablations, calibration analyses, and statistical benchmarks for foundation models, alongside LLM evaluation against physician judgment. Experienced building reproducible Python/SQL/PyTorch pipelines and translating ambiguous research questions into rigorous empirical studies.

EDUCATION

University of California, San Francisco (UCSF) — San Francisco, CA
M.S. in Health Data Science — Sept. 2026

University of California, Los Angeles (UCLA) — Los Angeles, CA
B.S. in Computational and Systems Biology, Concentration in Biological Data Sciences — Jun. 2024
Departmental Highest Honors

RESEARCH EXPERIENCE

UCSF Department of Medicine — San Francisco, CA
Data Science Specialist — Apr. 2025 – Apr. 2026

Empirical Evaluation of Large Language Models Against Human Expert Judgment

- Led statistical and analytical execution of an investigator-initiated study evaluating agreement between AI-generated and physician assessments of medical student OSCE performance.
- Consulted with clinical investigators to refine research questions, define evaluation criteria, and design validation analyses.
- Developed evaluation methodology using Cohen's κ, structured scoring, semantic similarity, and natural language inference to characterize agreement, reliability, and evaluator variability.
- Built reproducible Python workflows transforming multimodal clinical encounters into structured analytic data and evaluation outputs.
- Used physician feedback and model disagreement cases to refine evaluation criteria, study design, and subsequent analyses.
- Led IRB protocols, amendments, continuing reviews, grant submissions, and scientific outputs.

UCSF Hswen Lab — San Francisco, CA
Graduate Student Researcher — Oct. 2024 – Jun. 2026

Foundation Model Evaluation for Alzheimer's Disease Prediction

- Led research methodology for a multimodal study integrating 180,000 UK Biobank fundus images with demographic, ICD-10, and linked clinical data.
- Developed reproducible Python/SQL/PyTorch workflows for image QC, preprocessing, macular cropping, retinal vessel segmentation, and anatomical ablation.
- Trained and benchmarked RETFound Vision Transformers and CNNs against classical statistical baselines for Alzheimer's disease prediction.
- Designed repeated train/test evaluation, AUROC, calibration, statistical benchmarking, and controlled anatomical ablations to characterize robustness, variability, and information dependence.
- Investigated whether predictive performance reflected meaningful anatomical signal or artifacts of the dataset and experimental design, using additional experiments to distinguish competing explanations.
- Interpreted findings with collaborators across epidemiology, biostatistics, and neurology and communicated results through presentations and publication-quality visualizations.
- First-author work presented at AAIC 2026 and submitted as a first-author manuscript to Nature Medicine.

Behavioral and Systematic Analysis of Nicotine Pouch Use in Reddit

- Analyze 38,000+ Reddit posts to characterize behavioral patterns, sentiment, and public-health communication surrounding nicotine-pouch use.
- Develop reproducible NLP pipelines using text preprocessing, lemmatization, LDA topic modeling, and VADER sentiment analysis.
- Translate large-scale unstructured behavioral data into interpretable quantitative and qualitative research findings.

UCLA & PITTAN Los Angeles, CA
Undergraduate Honors Thesis Jun. 2023 – Jun. 2024
Sweat Amino Acid Analysis for Non-Invasive Skin Assessment

- Independently designed and executed an experimental study investigating sweat amino-acid biomarkers as a non-invasive diagnostic approach using 500+ clinical samples collected with Kobe University.
- Designed experimental workflows, coordinated international data collection, and performed statistical analysis of biochemical measurements.
- Developed and evaluated random forest and multimodal neural-network models integrating biochemical measurements for dermatological outcome prediction.
- Evaluated diagnostic performance, clinical feasibility, and translational potential relative to biopsy-based assessment.

UCLA Suthana Lab Santa Monica, CA
Research Assistant Mar. 2023 – Jun. 2024

- Processed and analyzed 1,000+ EEG recordings using MATLAB for clinical neuroscience research.
- Developed automated signal-processing and machine-learning pipelines to improve analytical efficiency, reproducibility, and data quality.
- Supported IRB-approved human-subjects research through participant coordination, consent documentation, and clinical study operations.

Institute of Neuro Innovation Santa Monica, CA
Data Analyst Aug. 2022 – Feb. 2023

- Analyzed clinical-trial data from 500+ participants evaluating virtual-reality audio therapy for depression.
- Designed and evaluated three sound-wavefront therapy protocols using statistical modeling to characterize treatment response.
- Managed research data and produced quantitative analyses supporting clinical research and treatment evaluation.

INDUSTRY EXPERIENCE

BOLERO Tokyo, Japan
Lead Data Scientist Apr. 2025 – Mar. 2026

- Led AI/ML and data strategy for production analytical systems supporting 100,000+ multilingual music tracks, translating ambiguous client and product needs into technical requirements and workflows.
- Architected end-to-end Python ML/data pipelines spanning ingestion, audio processing, feature extraction, representation learning, similarity analysis, and predictive modeling.
- Built web-based analytics and database infrastructure integrating audio, behavioral, time-series, metadata, and API data.
- Developed and evaluated LLM-based workflows for analytical and user-facing applications.
- Led implementation across ML, databases, APIs, web systems, automation, and security while troubleshooting production and data-quality issues.
- Directed and mentored technical team members across AI/data projects and collaborated across English- and Japanese-speaking stakeholders.

PITTAN Kobe, Japan
Chief Technology Officer Jun. 2023 – Sep. 2023

- Led technical development of a health-technology platform translating biochemical research into a user-facing diagnostic system.
- Designed Python/Raspberry Pi data-acquisition and workflow-automation systems for sweat analysis.
- Developed multimodal ML models integrating biochemical and sensor-derived biomarkers for skin-condition prediction.
- Coordinated technical implementation across research, engineering, and product functions in an international team.

SELECTED ABSTRACTS AND PUBLICATIONS

Hswen, Y., Han, E., **Yang, E.**, Ling,P., Hurley, M. (2026). "Understanding Real World Nicotine Pouch Narratives on Online Forums to Inform Responsive Health Communication." ***American Public Health Association (APHA 2026)*****,** San Antonio, TX. Nov. 2026

Yang, E., Buto, P., Vashisht, R., Hoffmann, T., Fu, C.-Y., La Joie, R., Brenowitz, W., Glymour, M., Torres, J., Hswen, Y. (2026). "Limits of Fundus-Based Deep Learning for Alzheimer's Disease Prediction: Anatomical Ablation and Clinical Benchmarking in UK Biobank." ***Alzheimer's Association International Conference (AAIC 2026)***, London, United Kingdom. July 2026

Yang, E., Buto, P., Vashisht, R., Hoffmann, T., Fu, C.-Y., La Joie, R., Brenowitz, W., Glymour, M., Torres, J., Hswen, Y. (2026). "Limits of Fundus-Based Deep Learning for Alzheimer's Disease Prediction: Anatomical Ablation and Clinical Benchmarking in UK Biobank." ***UCSF Department of Epidemiology and Biostatistics Research Day***, San Francisco, CA. May 2026

Yang, E. and Wong, F. (2026). "Simulating the Expert Evaluator: Immediate AI-Driven Feedback for Medical Student Oral Presentations and Written Documentation." ***UCSF Artificial Intelligence (AI) and Education Symposium***, San Francisco, CA. Feb. 2026.

Hswen, Y., **Yang, E.**, Han, E., Hurley, M., Criss, S., Nguyen, T., and Ling, P. (2026). "Themes of Online Discussions of Nicotine Pouch Use." ***Social Influence***. Under revision following peer review.

Yang, E. and Koyama, H. (2024). "Design a Skin Analysis Alternative Using Sweat Amino Acid Content for Commercial Use." **UCLA Quantitative Biosciences Research Symposium**, Los Angeles, CA; May 2024.

HONORS AND AWARDS

AAIC Conference Fellowship 2026
UCSF AGSD Summer Travel Award 2026

Skills & Interests

- **Programming:** Python, R, SQL, MATLAB, C/C++, Java, JavaScript
- **AI/ML:** PyTorch, LLMs, Deep Learning, Computer Vision, NLP, Multimodal Learning, Model Evaluation
- **Data & Analytics:** Statistical Analysis, Data Pipelines, APIs, Tableau, Power BI, Git/GitHub
- **Languages:** Mandarin Chinese (native), Japanese (intermediate)
- **Interests:** Percussion, Vocals, Digital Illustration, Creative Writing